Databricks ML Associate

Questions 801–900 (Ninth 100)

Practice Exam Deck

How to Use This Deck

100 new questions (Q801–Q900), none repeated from Q1–Q800.

Each question slide has four options (A–D).

The next slide gives the correct answer and explains why the right option is right and why each wrong option is wrong.

This set is **scenario-heavy**: you are given a situation and must choose the right Databricks tool, API, or workflow.

Question 801

A team wants to reuse the same customer features across three models and keep them governed in Unity Catalog. What should they build first?

A. A Feature Store table in Unity Catalog

B. A model serving endpoint

C. An AutoML experiment

D. A SQL warehouse

Answer 801

Correct: A — shared governed features belong in a Feature Store table.

Wrong:

B serves models, not features.

C trains models.

D runs SQL.

Question 802

A model was trained with `fe.log_model()`. At scoring time the input table has only `customer_id`. Which call should be used?

A. `fe.score_batch(model_uri, batch_df)`

B. `model.predict(batch_df)`

C. `mlflow.pyfunc.load_model(model_uri).predict(batch_df)`

D. `batch_df.join(feature_table)`

Answer 802

Correct: A — `score_batch()` uses logged feature metadata to join features.

Wrong:

B, C assume features are already present.

D is a plain join without automatic lookup.

Question 803

An offline model scores well but degrades online. Review shows a feature was computed using post-outcome data. What should have been used?

A. Point-in-time lookups in `create_training_set()`

B. A larger cluster

C. A different optimizer

D. More trees

Answer 803

Correct: A — point-in-time lookups prevent leakage.

Wrong:

B, C, D do not address leakage.

Question 804

A team wants one parent run for tuning and a child run per trial. What should they do?

A. Start child runs with `nested=True`

B. Turn on autologging only

C. Use the same experiment ID

D. Use a SQL warehouse

Answer 804

Correct: A — nested runs create parent-child relationships.

Wrong:

B does not create nesting.

C only groups runs in an experiment.

D is SQL compute.

Question 805

A data scientist wants to compare 50 hyperparameter runs visually and sort by RMSE. What should they use?

A. The MLflow experiment UI

B. A SQL warehouse

C. A notebook widget

D. A Repo

Answer 805

Correct: A — the experiment UI compares runs.

Wrong:

B, C, D do not compare runs.

Question 806

A team wants the run with the lowest RMSE in an experiment. Which call returns it?

A. `mlflow.search_runs(exp_id, order_by=["metrics.rmse ASC"])`

B. `mlflow.search_runs(exp_id, order_by=["metrics.rmse DESC"])`

C. `mlflow.best_run(exp_id)`

D. `mlflow.get_run(exp_id)`

Answer 806

Correct: A — ascending order puts the lowest RMSE first.

Wrong:

B returns the worst first.

C, D are not the API.

Question 807

A team wants a quick baseline across several algorithms without writing training code. What should they use?

A. Databricks AutoML

B. A SQL warehouse

C. A Repo

D. A cluster policy

Answer 807

Correct: A — AutoML builds baselines automatically.

Wrong:

B, C, D are not AutoML.

Question 808

AutoML produced a leaderboard. What must the team still do manually?

A. Deploy the best model

B. Tune hyperparameters

C. Evaluate models

D. Run EDA

Answer 808

Correct: A — deployment is outside AutoML.

Wrong:

B, C, D are automated.

Question 809

A team needs MLflow preinstalled and does not want to install libraries each time. What should they select?

A. A Databricks Runtime ML version

B. A SQL warehouse

C. A single-node cluster

D. A standard runtime without ML

Answer 809

Correct: A — Runtime ML includes MLflow.

Wrong:

B, C, D do not include ML libraries.

Question 810

A job has three dependent notebooks and fails intermittently. What is the best way to find the failing task?

A. Review the matrix view of the job run

B. Run each notebook interactively

C. Migrate to DLT

D. Increase cluster size

Answer 810

Correct: A — the matrix view shows task statuses.

Wrong:

B is debugging after identifying the task.

C, D do not diagnose failures.

Question 811

A team must improve a production notebook without changing what runs daily. What should they do?

A. Create a Git branch in Repos and commit changes there

B. Clone notebooks to a workspace folder

C. Create a new Git repo

D. Clone into a new Repo

Answer 811

Correct: A — branches isolate development.

Wrong:

B loses Git version control.

C, D are unnecessarily complex.

Question 812

A team wants scheduled retraining every day. What should they use?

A. A Databricks Job with a schedule trigger

B. A notebook widget

C. A SQL warehouse

D. A serving endpoint

Answer 812

Correct: A — Jobs support schedule triggers.

Wrong:

B, C, D do not schedule training.

Question 813

A team needs streaming inference as data arrives. What should they use?

A. Delta Live Tables

B. A SQL warehouse

C. A single-node cluster

D. A notebook widget

Answer 813

Correct: A — DLT supports streaming inference.

Wrong:

B, C, D are not streaming inference.

Question 814

A team needs low-latency REST predictions. What should they create?

A. A real-time serving endpoint
B. A batch inference job
C. A DLT pipeline
D. A SQL dashboard

Answer 814

Correct: A — real-time endpoints serve REST predictions.

Wrong:

B is offline.

C is streaming.

D is visualization.

Question 815

A model must score 100M Delta records overnight. What is the best approach?

A. Batch inference with a Spark UDF

B. A real-time endpoint per record

C. A notebook loop on the driver

D. Export to CSV and score externally

Answer 815

Correct: A — Spark UDF distributes inference.

Wrong:

B is high-latency per record.

C does not scale.

D is manual.

Question 816

A team wants to split traffic between two model versions on one endpoint. What should they configure?

A. Traffic splitting on the serving endpoint

B. Two separate endpoints

C. Two clusters

D. Two notebooks

Answer 816

Correct: A — traffic splitting routes requests.

Wrong:

B, C, D do not split traffic on one endpoint.

Question 817

A team wants to promote a model to production using aliases. What do they do?

A. Move the `champion` alias to the new version

B. Delete the old version

C. Retrain the model

D. Create a new experiment

Answer 817

Correct: A — promotion is moving the alias.

Wrong:

B deletes history.

C is unnecessary.

D is unrelated.

Question 818

A team wants to register a model from a run. Which call is correct?

A. `mlflow.register_model(f"runs:/{run_id}/model", "name")`

B. `mlflow.register_model(run_id, "name")`

C. `mlflow.register_model(f"runs:/{run_id}/model")`

D. `mlflow.register_model(f"runs:/{run_id}/name", "model")`

Answer 818

Correct: A — the URI includes the run and artifact path.

Wrong:

B is not a model URI.

C omits the name.

D swaps path and name.

Question 819

A team needs to view the notebook that created an MLflow run. What should they click?

A. The Source link on the experiment page

B. The MLmodel artifact

C. The Models link

D. The Start Time link

Answer 819

Correct: A — the Source link opens the notebook snapshot.

Wrong:

B is metadata.

C links to the registry.

D opens run details.

Question 820

A team wants the notebook to be reproducible months later. What is the best practice?

A. Use MLflow tracking for params, metrics, artifacts, and model
B. Print metrics to stdout
C. Save notebooks to local disk
D. Rely on cluster logs

Answer 820

Correct: A — MLflow tracking supports reproducibility.

Wrong:

B, C, D are not durable or centralized.

Question 821

A team needs a robust imputation for skewed numeric features. What should they use?

A. Median imputation

B. Mean imputation

C. Zero imputation

D. Drop rows

Answer 821

Correct: A — median is robust to skew.

Wrong:

B is distorted by skew.

C biases the distribution.

D loses data.

Question 822

A team must preserve information about missingness. What should they add?

A. A binary indicator feature per imputed column

B. A mean imputation

C. A constant feature

D. A drop of the column

Answer 822

Correct: A — indicator features preserve missingness.

Wrong:

B, C, D lose or distort information.

Question 823

A team wants to one-hot encode in Spark ML. What is the correct order?

A. StringIndexer then OneHotEncoder

B. OneHotEncoder then StringIndexer

C. VectorAssembler then OneHotEncoder

D. Imputer then OneHotEncoder

Answer 823

Correct: A — index first, then one-hot encode.

Wrong:

B is reversed.

C, D are unrelated.

Question 824

A data scientist builds a Spark ML `Imputer` and calls `transform` directly. Why does it fail?

A. The imputer was not fit first

B. The columns must be identical

C. It does not impute test data

D. The strategy is invalid

Answer 824

Correct: A — the imputer must be fit first.

Wrong:

B, C, D are false.

Question 825

A team wants to combine numeric and encoded categorical features. What should they use?

A. `VectorAssembler`

B. `StringIndexer`

C. `OneHotEncoder`

D. `Imputer`

Answer 825

Correct: A — VectorAssembler combines feature columns.

Wrong:

B, C, D are other transformers.

Question 826

A team wants a preprocessing and model pipeline fit inside CV. Where should the pipeline go?

A. Inside the cross-validator

B. Outside the cross-validator

C. As the final stage of itself

D. In a separate notebook

Answer 826

Correct: A — the pipeline belongs inside the CV to avoid leakage.

Wrong:

B causes leakage.

C is recursive.

D breaks integration.

Question 827

A team tunes 5 hyperparameters on 8 workers. Which approach best leverages the cluster?

A. Hyperopt with `SparkTrials` and TPE

B. Grid search on the driver

C. Random search on the driver

D. Manual tuning

Answer 827

Correct: A — Hyperopt + SparkTrials uses the cluster.

Wrong:

B, C, D are single-node or inefficient.

Question 828

With 3-fold CV and a 3x2 hyperparameter grid, how many models train in total?

A. 18

B. 6

C. 9

D. 36

Answer 828

Correct: A — $3 \times 2 \times 3 = 18$.

Wrong:

B is just combinations.

C, D are wrong arithmetic.

Question 829

A team wants fewer models trained because compute is limited. What should they choose?

A. A train-validation split

B. k-fold CV

C. Nested CV

D. Leave-one-out CV

Answer 829

Correct: A — a split trains fewer models.

Wrong:

B, C, D train more models.

Question 830

A fraud dataset has 1% positives. Which metric should be used?

A. PR-AUC or F1

B. Accuracy

C. RMSE

D. R-squared

Answer 830

Correct: A — imbalance-aware metrics are appropriate.

Wrong:

B is misleading.

C, D are regression metrics.

Question 831

A health team wants to catch as many positives as possible. Which metric should be optimized?

A. Recall

B. Precision

C. Accuracy

D. RMSE

Answer 831

Correct: A — recall maximizes positives caught.

Wrong:

B, C, D optimize other things.

Question 832

A model's RMSE is much higher than MAE. What does this suggest?

A. Large errors are present

B. Perfect predictions

C. No outliers

D. The model is underfit

Answer 832

Correct: A — RMSE penalizes large errors more.

Wrong:

B, C, D are not implied.

Question 833

A team evaluates a log(price) model against raw price labels and gets a huge RMSE. What was forgotten?

A. Exponentiating predictions before computing RMSE

B. Taking the log of labels

C. Scaling features

D. Imputing missing values

Answer 833

Correct: A — predictions must be transformed back.

Wrong:

B, C, D do not fix the scale mismatch.

Question 834

A team reports 3-fold CV RMSEs of 10, 12, 17. What is the overall CV RMSE?

A. 13.0

B. 17.0

C. 12.0

D. 39.0

Answer 834

Correct: A — $(10 + 12 + 17)/3 = 13.0$.

Wrong:

B, C, D are wrong arithmetic.

Question 835

Training accuracy is 99.5%, validation is 72%. What should be done?

A. Reduce complexity or add regularization

B. Increase depth

C. Deploy it

D. Add more test data

Answer 835

Correct: A — the gap indicates overfitting.

Wrong:

B worsens it.

C is unsafe.

D does not fix overfitting.

Question 836

A team wants to reduce variance without much extra bias. What should they use?

A. Bagging

B. Boosting

C. Increasing depth

D. Removing regularization

Answer 836

Correct: A — bagging reduces variance.

Wrong:

B reduces bias.

C, D increase variance.

Question 837

A team wants to improve bias sequentially. What should they use?

A. Boosting

B. Bagging

C. PCA

D. KMeans

Answer 837

Correct: A — boosting corrects errors sequentially.

Wrong:

B is parallel.

C, D are other methods.

Question 838

A team needs feature importance from a tree model. What should they inspect?

A. `featureImportances`

B. `coefficients`

C. `intercept`

D. `summary`

Answer 838

Correct: A — tree models expose `featureImportances`.

Wrong:

B, C are for linear models.

D is broad.

Question 839

A team wants to interpret a linear model. What should they inspect?

A. Coefficients

B. Feature importance

C. Trees

D. Clusters

Answer 839

Correct: A — linear models use coefficients.

Wrong:

B is for trees.

C, D are unrelated.

Question 840

A team wants to choose k for k-means. What should they use?

A. Elbow method or silhouette

B. ROC-AUC

C. RMSE

D. F1

Answer 840

Correct: A — elbow/silhouette select k.

Wrong:

B, C, D are other metrics.

Question 841

A team wants to reduce dimensionality before modeling. What should they use?

A. PCA

B. KMeans

C. ALS

D. StringIndexer

Answer 841

Correct: A — PCA reduces dimensions.

Wrong:

B, C are other algorithms.

D encodes strings.

Question 842

A team wants to cluster customers by behavior without labels. What should they use?

A. KMeans

B. LinearRegression

C. RandomForestClassifier

D. ALS

Answer 842

Correct: A — KMeans is unsupervised clustering.

Wrong:

B, C are supervised.

D is recommendation.

Question 843

A team wants recommendations from user-item interactions. What should they use?

A. ALS

B. KMeans

C. PCA

D. LinearRegression

Answer 843

Correct: A — ALS is collaborative filtering.

Wrong:

B, C, D are other methods.

Question 844

A team wants to detect rare anomalies without labels. What should they use?

A. Isolation Forest

B. LinearRegression

C. RandomForestClassifier

D. StringIndexer

Answer 844

Correct: A — Isolation Forest is unsupervised anomaly detection.

Wrong:

B, C are supervised.

D encodes strings.

Question 845

A team must serve features for real-time inference. What should they use?

A. An online feature table

B. An offline feature table

C. A SQL warehouse

D. A notebook

Answer 845

Correct: A — online tables serve low-latency features.

Wrong:

B is batch.

C, D are not feature stores.

Question 846

A team must train on historical features with point-in-time correctness. What should they use?

A. An offline feature table with timestamp keys

B. An online feature table

C. A SQL warehouse

D. A notebook

Answer 846

Correct: A — offline tables with timestamps support point-in-time joins.

Wrong:

B is for online serving.

C, D are unrelated.

Question 847

A team must share features across workspaces. What should they use?

A. Account-level Feature Store tables in Unity Catalog

B. Workspace-level tables

C. A notebook

D. A cluster

Answer 847

Correct: A — account-level tables are cross-workspace.

Wrong:

B is workspace-scoped.

C, D are unrelated.

Question 848

A team must update existing feature rows by key. Which write mode should they use?

A. `mode="merge"`

B. `mode="overwrite"`

C. `mode="append"`

D. `mode="replace"`

Answer 848

Correct: A — merge upserts by key.

Wrong:

B replaces all.

C appends.

D is not a mode.

Question 849

A team must ensure feature tables are available before training sets. What should they do?

A. Create feature tables first, then training sets

B. Create training sets first

C. Run them in parallel

D. Use only one

Answer 849

Correct: A — training sets depend on feature tables.

Wrong:

B is reversed.

C, D are invalid.

Question 850

A team needs cross-workspace governance for models. What should they use?

A. Unity Catalog model registry

B. Workspace model registry

C. A notebook

D. A cluster

Answer 850

Correct: A — Unity Catalog provides governance.

Wrong:

B is workspace-scoped.

C, D are unrelated.

Question 851

A team needs to use an alias in serving. What URI format should they use?

A. `models:/<name>@<alias>`

B. `models:/<alias>`

C. `runs:/<alias>`

D. `models:<name>`

Answer 851

Correct: A — alias URIs use `models:/<name>@<alias>`.

Wrong:

B, C, D are malformed.

Question 852

A team wants to set a model version description programmatically. Which call should they use?

A. `mlflow.update_model_version()`

B. `mlflow.set_tag()`

C. `mlflow.log_param()`

D. `mlflow.describe()`

Answer 852

Correct: A — it updates version metadata.

Wrong:

B, C set run metadata.

D is not the API.

Question 853

A team needs a description on a registered model. Which call should they use?

A. `mlflow.update_model()`

B. `mlflow.set_tag()`

C. `mlflow.log_artifact()`

D. `mlflow.register_model()`

Answer 853

Correct: A — `update_model()` sets model metadata.

Wrong:

B, C set run metadata.

D registers versions.

Question 854

A team needs an endpoint that scales to zero when idle. What should they configure?

A. Scale-to-zero on the endpoint

B. A larger cluster

C. A bigger model

D. A different runtime

Answer 854

Correct: A — scale-to-zero saves cost when idle.

Wrong:

B, C, D do not enable scale-to-zero.

Question 855

A team needs to authenticate serving calls. What should they use?

A. A personal access token or OAuth

B. Anonymous access

C. A shared password

D. A notebook

Answer 855

Correct: A — tokens authenticate serving calls.

Wrong:

B, C, D are insecure or unsupported.

Question 856

A team needs to monitor data drift in production. What should they track?

A. Input feature distributions over time

B. Training loss

C. Cluster CPU

D. Storage size

Answer 856

Correct: A — drift is a change in input distributions.

Wrong:

B, C, D are not drift metrics.

Question 857

A team needs to monitor concept drift. What should they track?

A. The feature-target relationship over time

B. Training loss

C. Cluster CPU

D. Storage size

Answer 857

Correct: A — concept drift changes the feature-target relationship.

Wrong:

B, C, D are unrelated.

Question 858

A team needs a model card. What should it include?

A. Intended use, data, performance, limitations

B. Model weights only

C. Cluster size

D. Notebook code

Answer 858

Correct: A — model cards document use and limitations.

Wrong:

B, C, D are incomplete.

Question 859

A team must satisfy an auditor about model decisions. What should they use?

A. SHAP explanations

B. A larger model

C. More data

D. A different optimizer

Answer 859

Correct: A — SHAP explains predictions.

Wrong:

B, C, D do not explain decisions.

Question 860

A team must detect bias in model outcomes. What should they do?

A. Run a bias audit across protected groups

B. Train more trees

C. Use a bigger cluster

D. Remove features randomly

Answer 860

Correct: A — bias audits assess fairness.

Wrong:

B, C, D do not assess bias.

Question 861

A team must protect PII in training data. What should they do?

A. Mask or pseudonymize identifiers

B. Log raw data

C. Share data publicly

D. Skip validation

Answer 861

Correct: A — masking/pseudonymization protects PII.

Wrong:

B, C, D are unsafe.

Question 862

A team must prevent a model from exposing PII in outputs. What should they do?

A. Validate and sanitize outputs before returning them

B. Train longer

C. Use a bigger cluster

D. Increase tree depth

Answer 862

Correct: A — output validation prevents exposure.

Wrong:

B, C, D do not prevent exposure.

Question 863

A team needs a secret for serving credentials. Where should they store it?

A. A Databricks secret scope

B. A notebook

C. A Git repo

D. A cluster log

Answer 863

Correct: A — secret scopes store secrets securely.

Wrong:

B, C, D expose secrets.

Question 864

A team needs to grant read access to a table. Which privilege should they grant?

A. `SELECT`

B. `MODIFY`

C. `EXECUTE`

D. `OWN`

Answer 864

Correct: A — `SELECT` grants read.

Wrong:

B, C, D are other privileges.

Question 865

A team needs to grant write access to a table. Which privilege should they grant?

A. `MODIFY`

B. `SELECT`

C. `EXECUTE`

D. `READ`

Answer 865

Correct: A — `MODIFY` grants write.

Wrong:

B, C, D are other privileges.

Question 866

A team needs to grant permission to serve a model. Which privilege should they grant?

A. `EXECUTE`

B. `SELECT`

C. `MODIFY`

D. `READ`

Answer 866

Correct: A — `EXECUTE` allows model use.

Wrong:

B, C, D are table privileges.

Question 867

A team needs to limit what rows a user can see. What should they use?

A. A row filter

B. A column mask

C. A cluster policy

D. A secret scope

Answer 867

Correct: A — row filters limit visible rows.

Wrong:

B masks columns.

C constrains clusters.

D stores secrets.

Question 868

A team needs to hide sensitive column values from some users. What should they use?

A. A column mask

B. A row filter

C. A cluster policy

D. A secret scope

Answer 868

Correct: A — column masks hide values.

Wrong:

B filters rows.

C, D are unrelated.

Question 869

A team needs lineage from a table to a model. What should they use?

A. Unity Catalog lineage

B. A notebook

C. A cluster

D. A secret scope

Answer 869

Correct: A — UC lineage tracks dependencies.

Wrong:

B, C, D do not provide lineage.

Question 870

A team needs a governed path for external files. What should they use?

A. An external location

B. A notebook

C. A cluster

D. A secret scope

Answer 870

Correct: A — external locations govern file paths.

Wrong:

B, C, D are unrelated.

Question 871

A team needs to manage cloud credentials for UC storage. What should they use?

A. A storage credential

B. A notebook

C. A cluster

D. A secret scope

Answer 871

Correct: A — storage credentials manage cloud access.

Wrong:

B, C, D are unrelated.

Question 872

A team needs to share data with another organization. What should they use?

A. Delta Sharing
B. A notebook
C. A cluster
D. A secret scope

Answer 872

Correct: A — Delta Sharing shares data securely.

Wrong:

B, C, D do not enable sharing.

Question 873

A team needs to automate deployment of jobs and pipelines. What should they use?

A. Databricks Asset Bundles

B. A notebook

C. A cluster

D. A secret scope

Answer 873

Correct: A — bundles automate deployment.

Wrong:

B, C, D do not automate deployment.

Question 874

A team needs to sync notebooks with Git. What should they use?

A. Databricks Repos

B. A secret scope

C. A cluster policy

D. A SQL warehouse

Answer 874

Correct: A — Repos sync with Git.

Wrong:

B, C, D are unrelated.

Question 875

A team needs data quality rules enforced in a pipeline. What should they use?

A. Delta Live Tables expectations

B. A notebook

C. A cluster

D. A secret scope

Answer 875

Correct: A — DLT expectations enforce quality rules.

Wrong:

B, C, D do not enforce rules.

Question 876

A team needs to drop bad records during a DLT run. Which expectation should they use?

A. `expect_or_drop`

B. `expect_or_fail`

C. `expect_all`

D. `expect_none`

Answer 876

Correct: A — `expect_or_drop` drops bad records.

Wrong:

B fails the pipeline.

C, D are not valid.

Question 877

A team needs to fail a DLT pipeline on data quality violations. Which expectation should they use?

A. `expect_or_fail`

B. `expect_or_drop`

C. `expect_all`

D. `expect_none`

Answer 877

Correct: A — `expect_or_fail` fails the pipeline.

Wrong:

B drops records.

C, D are not valid.

Question 878

A team needs to add columns to a Delta table without rewriting data. What should they use?

A. Schema evolution with `mergeSchema`

B. `overwriteSchema`

C. A notebook

D. A cluster

Answer 878

Correct: A — `mergeSchema` evolves schemas.

Wrong:

B replaces the schema.

C, D are unrelated.

Question 879

A team needs to upsert into a Delta table. What should they use?

A. `MERGE INTO`

B. `INSERT INTO`

C. `CREATE TABLE`

D. `DROP TABLE`

Answer 879

Correct: A — `MERGE INTO` upserts.

Wrong:

B appends.

C creates.

D deletes.

Question 880

A team needs to compact small files in a Delta table. What should they run?

A. `OPTIMIZE`

B. `VACUUM`

C. `MERGE`

D. `DROP`

Answer 880

Correct: A — OPTIMIZE compacts files.

Wrong:

B removes old files.

C upserts.

D deletes.

Question 881

A team needs to remove stale files not needed for time travel. What should they run?

A. `VACUUM`

B. `OPTIMIZE`

C. `MERGE`

D. `DROP`

Answer 881

Correct: A — VACUUM removes stale files.

Wrong:

B compacts files.

C upserts.

D deletes the table.

Question 882

A team needs to query a previous version of a Delta table. What should they use?

A. Time travel

B. OPTIMIZE

C. VACUUM

D. MERGE

Answer 882

Correct: A — time travel queries old versions.

Wrong:

B, C, D are write/maintenance operations.

Question 883

A team needs a data profile before modeling. What should they run?

A. `df.summary()`

B. `df.show()`

C. `df.printSchema()`

D. `df.count()`

Answer 883

Correct: A — `summary()` returns descriptive stats.

Wrong:

B, C, D do not give full stats.

Question 884

A team needs to filter rows where `price > 0`. Which code works?

A. `df.filter(col("price") > 0)`

B. `df[df["price"] > 0]`

C. `df.loc[df["price"] > 0]`

D. `df.query("price > 0")`

Answer 884

Correct: A — `filter(col(...))` is the PySpark way.

Wrong:

B, C, D are pandas-style.

Question 885

A team needs a Spark DataFrame summary of count, mean, std, min, max, and quartiles. Which call should they use?

A. `df.summary()`

B. `df.describe().head()`

C. `df.stats()`

D. `df.printSchema()`

Answer 885

Correct: A — `summary()` includes quartiles.

Wrong:

B lacks quartiles.

C, D are not stats APIs.

Question 886

A team needs an outlier filter by IQR. Which range should they use?

A. $[Q1 - 1.5IQR,\ Q3 + 1.5IQR]$

B. $[Q1 - IQR,\ Q3 + IQR]$

C. $[Q1 - 2IQR,\ Q3 + 2IQR]$

D. $[Q1,\ Q3]$

Answer 886

Correct: A — the $1.5\times$IQR rule is standard.

Wrong:

B, C, D use wrong thresholds.

Question 887

A team needs to log a run to a specific experiment. Which call should they use?

A. `mlflow.set_experiment()`

B. `mlflow.start_run()`

C. `mlflow.log_artifact()`

D. `mlflow.register_model()`

Answer 887

Correct: A — `set_experiment()` sets the active experiment.

Wrong:

B starts a run.

C logs files.

D registers models.

Question 888

A team needs to log a dictionary as an artifact. Which call should they use?

A. `mlflow.log_dict()`

B. `mlflow.log_metric()`

C. `mlflow.log_param()`

D. `mlflow.set_tag()`

Answer 888

Correct: A — `log_dict()` stores JSON artifacts.

Wrong:

B, C, D log other things.

Question 889

A team needs to log a figure. Which call should they use?

A. `mlflow.log_figure()`

B. `mlflow.log_metric()`

C. `mlflow.log_param()`

D. `mlflow.set_tag()`

Answer 889

Correct: A — `log_figure()` logs figures.

Wrong:

B, C, D log other things.

Question 890

A team needs to log a model artifact. Which call should they use?

A. `mlflow.log_artifact()`

B. `mlflow.log_param()`

C. `mlflow.log_metric()`

D. `mlflow.set_tag()`

Answer 890

Correct: A — `log_artifact()` logs files.

Wrong:

B, C, D log other things.

Question 891

A team needs to log a model in scikit-learn flavor. Which call should they use?

A. `mlflow.sklearn.log_model()`

B. `mlflow.log_model()`

C. `mlflow.sklearn.save()`

D. `mlflow.sklearn.register()`

Answer 891

Correct: A — `sklearn.log_model()` logs the flavor.

Wrong:

B is generic.

C, D are not the API.

Question 892

A team needs to log a Spark ML model. Which flavor should they use?

A. `mlflow.spark`

B. `mlflow.sklearn`

C. `mlflow.pytorch`

D. `mlflow.tensorflow`

Answer 892

Correct: A — Spark ML uses the `spark` flavor.

Wrong:

B, C, D are other flavors.

Question 893

A team needs a custom prediction wrapper. What should they use?

A. A custom `pyfunc` model

B. A sklearn model

C. A Spark ML model

D. A PyTorch model

Answer 893

Correct: A — pyfunc wraps custom prediction logic.

Wrong:

B, C, D are standard flavors.

Question 894

A team needs to validate model inputs at serving. What should they add?

A. A model signature

B. A model alias

C. A model description

D. A model tag

Answer 894

Correct: A — signatures validate schemas.

Wrong:

B, C, D are metadata.

Question 895

A team needs to review input examples at serving. What should they log with the model?

A. An input example

B. A model tag

C. A model description

D. A model alias

Answer 895

Correct: A — input examples document expected inputs.

Wrong:

B, C, D are metadata.

Question 896

A team needs to trace a model version to its originating run. What should they inspect?

A. The source run link

B. The model alias

C. The model description

D. The model tag

Answer 896

Correct: A — source run links provide traceability.

Wrong:

B, C, D are metadata.

Question 897

A team needs to archive an old model version. What should they do?

A. Transition it to Archived or delete it

B. Rename it

C. Retrain it

D. Serve it

Answer 897

Correct: A — archiving retires versions.

Wrong:

B, C, D do not retire it.

Question 898

A team needs to compare two models in production. What should they use?

A. Traffic splitting on one endpoint

B. Two clusters

C. Two notebooks

D. Two feature tables

Answer 898

Correct: A — traffic splitting enables comparisons.

Wrong:

B, C, D do not split traffic.

Question 899

A team needs an automated release of a new model version. What should they use?

A. CI/CD with MLflow and Repos/Bundles

B. A notebook

C. A cluster

D. A secret scope

Answer 899

Correct: A — CI/CD automates releases.

Wrong:

B, C, D do not automate releases.

Question 900

A team needs to reproduce a model months later. What is the best practice?

A. Log params, metrics, artifacts, model, and environment

B. Save only the model weights

C. Print metrics to stdout

D. Use a bigger cluster

Answer 900

Correct: A — full MLflow tracking supports reproducibility.

Wrong:

B, C, D are incomplete or not durable.